

REVENUE LEAKAGE AND OPERATIONAL RISK ANALYSIS

UK online retail sales, December 2009 to December 2011

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1. INTRODUCTION

This report explains what I found in two Power BI dashboards that I built on UK online retail sales data. The data runs from December 2009 to 9 December 2011 and has about 1.03 million transaction lines from 53,628 invoices. The first dashboard is the Revenue Leakage Dashboard. It shows how much money is lost between gross sales and net revenue. The second is the Performance and Operational Risk Dashboard. It looks at how many invoices are completed, how customers behave by RFM score, and where stock adjustments happen.

I went through every visual on both pages. I used the month filter, the Top N filter (5, 10, 15 and 20) and the click filters between the visuals, and then I checked the numbers again against the data model behind the dashboards. In a few places this check changed my first reading of a chart, and I have explained those places clearly.

Main findings in short

1. The leakage rate is **7.14%**, but two orders that were bought and cancelled on the same day (£168K and £77K) pull it up. Without them, the rate is **6.02%**.
2. **51%** of the leakage is not customer cancellation. It is fees and adjustments, mainly the codes M and AMAZONFEE.
3. Product cancellations are £716K, which is 3.65% of product sales. After removing the two reversed orders, it is 2.43%.
4. The invoice completion rate of 74.74% looks low because about 10% of all invoices are stock adjustments, not customer orders. Counting only customer orders, the rate is 82.9%.
5. The stock loss of 805K units counts added stock and removed stock both as loss. The net removal is 333K units.
6. The UK has 86.6% of the leakage, but its rate (7.27%) is close to the rest of the countries (6.41%). The very high rates are in small markets such as Singapore and Hong Kong.

2. Data and definitions

The data is the Online Retail II dataset. It is a real record of the transactions of a UK-based online shop that mainly sells giftware, and many of its customers are wholesalers. The original data has 1,067,371 rows in two yearly sheets (2009 to 2010 and 2010 to 2011), which I combined in Power Query. I removed rows that were fully identical (same invoice, stock code, description, quantity, date, price, customer, and country). The final table has 1,033,036 rows, so 34,335 rows were taken out, mostly by

this duplicate step. Two summary tables, RFM_Customers and Product_Summary, were prepared in Excel and loaded into the same model.

About 235,000 rows (23% of the table) have no customer ID. Their gross sales are £3.10M and their leakage is £378K. This matters because every customer level chart leaves these rows out, so such charts show only about three quarters of the leakage.

Every row gets a transaction type, and all the numbers in this report depend on these rules, so I list them in Table 1.

Table 1. How the main terms are defined in the dashboards

Term	Definition used
Cancelled	The invoice number starts with the letter C (a credit note).
I/A (inventory adjustment)	Price is 0 and there is no customer ID. The descriptions are things like check, damaged, found and missing.
Zero-priced item	The price is 0 but a customer ID is present. Only 70 rows.
Completed	All the other rows.
Gross sales	Quantity times price for completed rows with a positive amount.
Leakage value	Quantity times price of cancelled rows, taken as a positive number. In Power Query, it is called cancellation value.
Net revenue	Gross sales minus leakage value.
Leakage rate	Leakage value is divided by gross sales.
Item type	Non-product for 15 codes (M, AMAZONFEE, POST, BANK CHARGES, ADJUST, ADJUST2, CRUK, D, S, B, C2, C3, DOT, TEST001, TEST002) and anything starting with GIFT. Product for everything else.
Completion rate of invoice	Invoices with at least one completed row divided by all invoices.
Loss-to-sale %	Inventory adjustment quantity is divided by the total quantity of the product (sold plus cancelled plus adjusted).
RFM segment	Based on the total of the R, F, and M scores (each score is 1 to 5, and the totals are 3 to 14 in this data). 13 and above are Champion, 10 to 12 Strong, 7 to 9 Developing and below 7 Low Engagement.

3. Dashboard 1: Revenue Leakage

3.1 The overall picture

Table 2. Headline numbers, all months and all countries

Measure	Value	Measure	Value
Gross sales	£20,476,260	Cancelled invoices	8,292
Leakage value	£1,462,798	Countries affected	36
Net revenue	£19,013,463	Customers affected	2,572
Leakage rate	7.14%	Products affected	2,886

The revenue bridge on the dashboard goes from £20.48M of gross sales to £19.01M of net revenue after £1.46M of leakage. A rate of 7.14% sounds high for retailers. The next sections show why this number needs careful reading.

3.2 Two orders that were reversed on the same day

When I clicked customer 16446 in the Top N Customers chart, the whole page changed to one invoice, one product (23843) and one country. Gross sales were £168.47K and leakage was £168.48K, so the rate was 100% and the net revenue was almost zero. In the data, this customer bought 80,995 units of product 23843 on invoice 581483 at 09:15 on 9 December 2011. The same quantity was cancelled on invoice C581484 at 09:27, twelve minutes later.

Customer 12346 did the same on 18 January 2011. They bought 74,215 units of product 23166 (£77.18K) and cancelled the order sixteen minutes later. I cannot say from the data why this happened. It looks like an order that was entered and then reversed straight away. Since many customers of this shop are wholesalers, a big order is not strange by itself, but a full cancellation within minutes is.

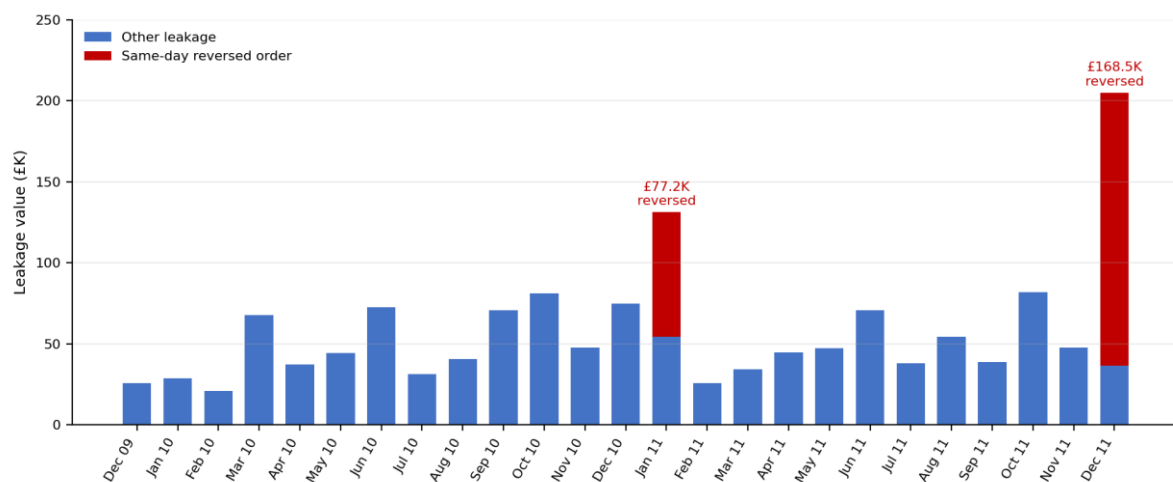
Together, the two pairs are £245.65K, which is 16.8% of all the leakage in the two years. No real revenue was lost, because the sales and the cancellation canceled each other. But both pairs are inside the leakage numbers, and they make January 2011 and December 2011 look bad months. Table 3 shows the effect.

Table 3. Leakage rate as shown in the dashboard and without the two reversed orders

Measure	As shown	Without the two orders
Leakage rate, all data	7.14%	6.02%
Leakage value, all data	£1,462.80K	£1,217.14K
January 2011 rate	19.0%	8.8%
December 2011 rate	32.2%	7.8%
January, all years (month filter = Jan)	11.93%	6.56%
December, all years (month filter = Dec)	13.40%	6.49%
Year 2010 rate	6.28%	6.28%
Year 2011 rate	8.34%	5.99%

The last two rows are the most useful for me. As shown in the dashboard, 2011 looks worse than 2010 (8.34% against 6.28%). After taking out the two reversed orders, 2011 is slightly better (5.99%). So, the apparent rise is not a real rise. Figure 1 shows the monthly leakage with the reversed parts in red.

Figure 1. Monthly leakage value from December 2009 to December 2011. The red parts are the two same-day reversed orders.

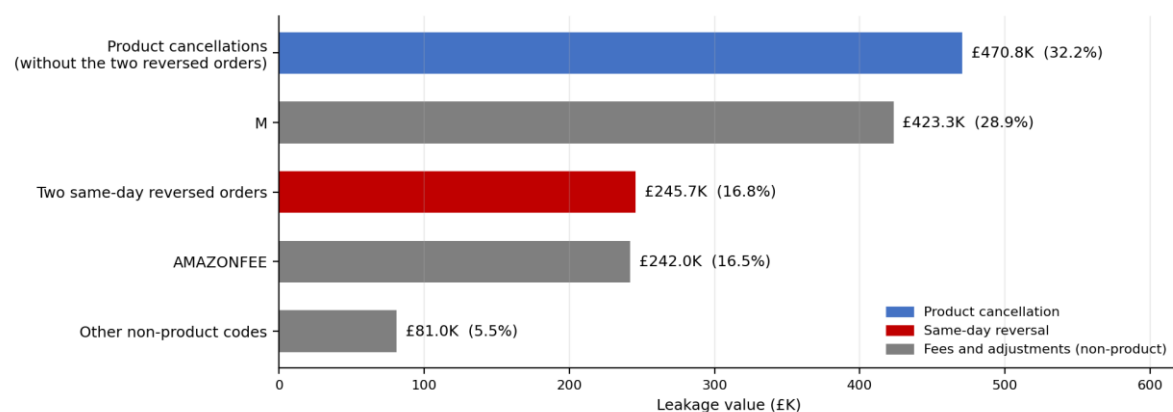


3.3 Half of the leakage is fees and adjustments

The Non-Product Leakage Sources chart shows the codes that are not real products. They add up to £746.3K, which is 51.0% of all the leakage. The two biggest are M (£423.3K) and AMAZONFEE (£242.0K). Bank charges are £36.0K and postage is £15.3K. These are not customers cancelling goods. AMAZONFEE has only £20.5K in gross sales but £242.0K in leakage, so the fees are recorded as credit notes, and the dashboard counts them as cancellations. In simple words, the leakage gauge mixes two different things.

Figure 2 splits the £1.46M into its parts. Real product cancellations, without the two reversed orders, are £470.8K, which is only 32.2% of the leakage. On all product sales (£19.64M), product cancellations are £716.5K or 3.65%, and 2.43% after removing the two reversed orders.

Figure 2. Where the £1.46M of leakage comes from.



As far as I understand, the codes M, POST, D, S, C2, CRUK and DOT stand for manual entries, postage, discount, samples, carriage, CRUK commission and Dotcom postage. I could not confirm this from the data alone, so it should be checked with the owner of the data before the numbers are used for any decision.

3.4 Month by month

The Monthly Leakage Trend shows the value of bars and the rate as a line. The value follows the sales. Once the two reversed orders are set aside, October is the highest month in both years (£81.3K in 2010 and £81.9K in 2011), when customers buy before Christmas. The rate gives a different picture. In both years November has the highest sales but the lowest rate (3.3% in 2010 and 3.2% in 2011). June is the highest rate in both years (9.7% and 9.3%), and December 2010 is 9.1%. All twelve months are in Appendix A.

Two points to remember when reading the trend. The chart starts in January 2010, so the £25.8K from December 2009 (£25,835 in my Excel pivot) is in total but not in the bars (the bars add to £1,436.96K). Also, December 2011 has only nine days of data because the file ends on 9 December, so it cannot be compared with a full December.

3.5 Countries

The United Kingdom has £1,266.2K of the leakage, which is 86.6% of the total. Its rate of 7.27% is close to the 6.41% of all other countries together. So, the UK is a big contributor mainly because of its size.

Table 4. Leakage by country, selected countries

Country	Gross sales	Leakage	Leakage rate	Invoices
United Kingdom	£17,410.2K	£1,266.2K	7.27%	49,108
EIRE	£658.8K	£48.8K	7.41%	806
France	£350.5K	£28.7K	8.20%	746
Norway	£56.3K	£20.9K	37.0%	53
Spain	£108.3K	£17.3K	16.0%	188
Germany	£425.0K	£13.1K	3.07%	1,095
Singapore	£25.3K	£12.2K	48.0%	14
Hong Kong	£23.7K	£9.9K	41.6%	21

Some small markets have very high rates, such as Singapore (48.0%), Hong Kong (41.6%) and Malta (35.9%, £2.9K on £8.1K of sales). They have only 14 to 21 invoices each, so one cancelled order can change the rate a lot. I would not call these countries risky without more data. Norway (37.0%) and Spain (16.0%) are more worth a look because their sales are bigger (£56.3K and £108.3K, with 53 and 188 invoices).

3.6 Customers and products

The leakage is concentrated on a few customers. The top 3 customers have 19.5% of all the leakage, the top 5 have 23.2%, and the top 10 have 31.2%.

Table 5. Top five customers by leakage value

Customer	Leakage	Rate	What I saw
16446	£168.5K	100%	Bought and cancelled product 23843 on 9 December 2011, twelve minutes apart.
12346	£77.6K	100%	Bought and cancelled product 23166 on 18 January 2011, sixteen minutes apart.
15098	£39.3K	98%	Cancelled almost everything that was bought.
16029	£28.1K	24%	Large buyer (£117.8K of sales), so the rate is lower.
14063	£26.5K	117%	Cancellations are bigger than sales in the data. The original purchase is probably before December 2009.

The customer chart also leaves rows with no customer ID. Those rows hold £378.0K of the leakage (25.8%), so the customer view shows only about three quarters of it.

For products, the top five have 19.1% of the leakage. Two of them are the same reversed orders (23843 and 23166). The other three are normal products: 22423 REGENCY CAKESTAND 3 TIER (£16.5K), 85123A WHITE HANGING HEART T-LIGHT HOLDER (£9.4K) and 71477 COLOUR GLASS STAR T-LIGHT HOLDER (£7.2K). Their rates are 5%, 4%, and 16%, which is not alarming. The first two have high sales (£330.6K and £261.2K), so their leakage comes mostly from volume.

4. Dashboard 2: Performance and Operational Risk

4.1 Invoice completion rate

The gauge shows 74.74%, which is 40,082 completed invoices out of 53,628. This looks low, but the total includes invoices that are not customer orders. Table 6 splits 53,628 invoices into groups.

Table 6. Invoices by group

Invoice group	Invoices	Share
Completed (customer orders)	40,082	74.74%
Cancelled (credit notes)	8,292	15.46%
Stock adjustment or zero-price only	5,254	9.80%
Total	53,628	100.00%

Stock adjustments have no customer ID, so they are not orders at all. If I leave them out, the completion rate is **82.9%** (40,082 out of 48,374). In the latest version of the file, page 2 has a Product only filter. With that filter the rate is 75.77% (39,516 out of 52,155 invoices).

One limit to keep in mind. A cancellation is a separate credit note, and the original sale invoice stays as completed. The dashboard does not link the two. So an order that was cancelled later is still counted as a completed invoice, and the completion rate is probably a little high for that reason.

4.2 Completion by country

The UK has 36,540 of the 40,082 completed invoices (91.2%). Its completion rate is 74.4%, and because it is so large, it pulls the overall rate down. Smaller countries look better. The Netherlands is at 91.2% and France at 83.4%. Germany has the lowest rate among the large markets (72.1% on 1,095 invoices).

Table 7. Invoice completion for the largest countries

Country	Invoices	Completed	Completion rate
United Kingdom	49,108	36,540	74.4%
Germany	1,095	789	72.1%
EIRE	806	626	77.7%
France	746	622	83.4%
Netherlands	250	228	91.2%
Spain	188	154	81.9%

On the dashboard, the Top 10 Countries chart is sorted by rate and not by volume, which is why the UK comes near the end. The UK bar is also many times bigger than the others, so the other countries are almost invisible. A table, or a log scale, would read better.

4.3 Customer RFM and invoice risk

The RFM scores were calculated in Excel. Recency, frequency and net spending are each split at the 20th, 40th, 60th and 80th percentiles to give a score from 1 to 5, and a recent buyer gets a higher recency score. Recency is counted in days up to 10 December 2011.

Table 8. RFM segments (5,878 customers with a customer ID)

Segment	Customers	Share of net spending	Average completion per customer	Leakage rate
Low Engagement (3 to 6)	941	0.8%	91.7%	58.5%
Developing (7 to 9)	2,810	23.2%	88.3%	8.9%
Strong (10 to 12)	2,096	55.7%	86.7%	3.5%
Champion (13 and above)	31	20.2%	82.5%	3.7%

The scatter plot looks like Champions have the worst completion, but I think this is misleading. Low Engagement customers have only 1.5 invoices each on average (1,396 invoices for 941 customers), so one cancelled invoice moves their rate a lot. If I weigh invoices, all four segments are close to each other (81.5% to 83.4%). The differences in the table are mostly because of the number of invoices per customer, and not because the segments behave differently.

The leakage rate is different. Low Engagement customers have 58.5% (£191.2K on £327.0K), against 3.5% to 3.7% for Strong and Champion customers. Possibly many of them bought once and then cancelled. They are only 0.8% of net spending, so the effect on the business is small.

Revenue is concentrated. The 2,127 Strong and Champion customers (36% of customers) bring 75.9% of net spending. The 31 Champions alone bring 20.2% (£3.31M). The RFM table covers £1.02M of the leakage, because rows with no customer ID are not in it.

4.4 Inventory adjustment (stock loss)

The Quantity I/A card shows 805K units against about 11M units sold. I need to explain what this number really is. An inventory adjustment row has price 0 and no customer ID. There are 5,944 such

rows on 5,318 invoices. The descriptions are things like check (163 rows), damaged (99), a question mark (90), damages (88), found (38) and missing (30).

The dashboard takes the absolute value of every row, so stock that was found or added back is also counted as loss. Of the 805,320 units, 569,314 were removed from stock and 236,006 were added. The net removal is **333,308 units**, which is 3.0% of units sold, against 7.2% if I use the absolute number.

Table 9. Top products on the Inventory Loss chart

Product	I/A units (dashboard)	Net units	Rows	Loss-to-sale %
84016	38,895	-2,903	11	97.3%
22759	19,640	-214	5	67.6%
23005	19,200	-19,200	2	80.2%
20852	18,800	-18,800	5	100%
22126	16,400	-16,400	3	100%
22752	12,000	0	2	76.5%
22753 to 22758 (each)	10,000	0	2	71.8% to 89.2%

Several of these are not real losses. Products 22752 to 22758 each have two adjustment rows that net to zero. Stock was added on 5 August 2010 and removed again on 13 August 2010, so it looks like an entry and its correction, but the chart shows 10,000 to 12,000 units of loss for each. On the other hand, 20852, 22126, and 23005 are real removals. The first two have no sales at all, which is why their loss-to-sale rate is 100%. Product 84016 shows 38,895 units in the dashboard but only 2,903 were removed in net terms.

4.5 Quantity-based product performance

This scatter plot compares the quantity sold with the quantity cancellation rate. Most products sit at a low end. The median rate is 0.2% and 75% of the products are below 1.6%. The two large bubbles near 80,000 units are products 23843 and 23166, the same reversed orders. They show about 50% and 49% and not 100%, because the rate divides the cancelled quantity by sold plus cancelled units. The Dynamic Product Segment puts both of them in Popular High Risk, which is correct by the rules but not useful in practice, since these were not normal sales.

5. Problems I found in the dashboards

Table 10 lists the problems, why each one matters, and how I plan to fix them. The first three changed the main message of the dashboards, so I would fix them first.

Table 10. Problems found and suggested fixes

No.	Problem	Why it matters	Suggested fix
1	Two same-day reversed orders (£245.7K)	They lift the leakage rate from 6.02% to 7.14% and make January and December 2011 look bad.	Add a flag for orders cancelled on the same day. Show the rate with and without them.
2	Fees and adjustments counted as cancellations	51% of the leakage is not customer cancellation.	Show product cancellation rate and fees and adjustments as separate cards. The Item_Type column already exists.

No.	Problem	Why it matters	Suggested fix
3	Labels still say cancellation	The bridge stage, tooltips and card names mix the words cancellation and leakage.	Use leakage for the total and cancellation only for product cancellations.
4	December 2009 missing in the trend chart	The bars add to £1,436.96K, the headline is £1,462.80K.	Include 2009 or add a footnote.
5	December 2011 has nine days only	It cannot be compared with a full month.	Mark it as a partial month.
6	Month filter combines the years	Dec shows 13.40%, which mixes three different Decembers.	Add a year slicer or label the filter for all years.
7	Rates above 100% and rates on tiny bases	Customer 14063 shows 117%. Singapore shows 48% on 14 invoices.	Explain or cap at 100%. Add a minimum sale or invoice count for countries.
8	Customer view skips blank customer IDs	25.8% of the leakage is not in the customer chart.	Add an unknown customer bar, or a note on the chart.
9	Completion rate includes stock adjustments	9.8% of the invoices are not orders.	Remove them from the denominator, or rename the measure.
10	Inventory loss uses absolute values	Added stock counts as loss. 805K units against 333K net.	Show removed, added and net separately.
11	Product_Summary has 172 stock codes twice	For example, 15056BL has one row with 116 units and another with 9,807. The visuals sum up a ratio column, so a duplicated code may add two percentages together. My guess is a difference in upper and lower cases, but I have not confirmed it.	Clean the codes in Power Query. Use a DIVIDE measure instead of summing a ratio column.
12	Top 10 products chart shows 13 bars	Six products ties at 10,000 units and Top N keep ties.	Add a tiebreaker such as total quantity.
13	Gauge scale	The gauge is green with up to 25% leakage, so 7% always looks safe.	Use a tighter target, for example, amber above 5%.
14	Spelling and file paths	The segment's name is spelled Champion. RFM and Product_Summary load from a local Downloads folder, so refreshing will fail on another computer.	Correct the spelling. Keep the Excel file in a shared folder.

6. Recommendations

1. **Show two rates side by side.** The leakage rate (7.14%) and the product cancellation rate (3.65%, or 2.43% without the reversed orders). Then it is clear what a real cancellation is.
2. **Ask the data owner what M and AMAZONFEE really are.** Together they are £665K, which is 45% of all the leakage. They may belong to marketplace fees or manual entries and not to revenue leakage.
3. **Watch for large orders cancelled on the same day.** A simple alert would have caught customers 16446 and 12346 early.

4. **Fix the stock adjustment measure and the reason codes.** Show removed, added, and net units. Many descriptions are vague (check, a question mark), so a proper reason code would make this measure much more useful.
5. **Clean the dashboards using Table 10,** mainly the duplicate stock codes in Product_Summary and a minimum size for country rates.
6. **Use net numbers when talking about performance.** Net revenue is £19.01M, and after taking out the reversed orders the leakage rate did not rise from 2010 to 2011.

7. Limitations

This analysis has been using the data of one retailer for about two years. There is no cost or margin data, so I can measure lost sales value but not lost profit. The data does not say why an invoice was cancelled, so I cannot separate customer returns from mistakes. My reading of the stock codes and adjustment descriptions comes from the text in the data and should be confirmed with the business. A cancellation is not linked to its original invoice, so I cannot say how many completed invoices were cancelled later. Finally, December 2011 is incomplete, as the data stops on 9 December.

Appendix A: Leakage by calendar month

Each row combines all the years in the data for that month, which is what the month filter does. December therefore includes December 2009, 2010, and 2011.

Table A1. Leakage value and rate by month, as shown and without the two reversed orders

Month	Leakage (£K)	Rate as shown	Rate without the two orders
January	160.04	11.93%	6.56%
February	46.51	4.33%	4.33%
March	101.87	6.58%	6.58%
April	81.95	6.74%	6.74%
May	91.64	6.42%	6.42%
June	143.03	9.47%	9.47%
July	69.36	5.07%	5.07%
August	94.81	6.52%	6.52%
September	109.43	5.53%	5.53%
October	163.19	7.05%	7.05%
November	95.32	3.21%	3.21%
December	305.65	13.40%	6.49%
Total	1,462.80	7.14%	6.02%

Appendix B: Filter views I checked

These are the filter views from the dashboards that I used to read the numbers. The sales, leakage, rates, and the counts of invoices, countries and customers agree with the data model. The products affected card counts real products only, because it has the Product filter. That is why it shows 2,886 for all months, while all stock codes with a cancellation would be 2,898.

Table B1. Filter views and what they show

Filter view	What it shows
Month = Jan	Leakage £160.04K, rate 11.93%, 560 cancelled invoices, 15 countries, 386 customers, 726 products.
Month = Dec	Gross £2,281.75K, leakage £305.65K, net £1,976.09K, rate 13.40%, 1,010 products affected.
Click customer 16446	2 cancelled invoices, 1 country, 1 product (23843), rate 100%. The second invoice is a test entry of £9.
Click product 23843	1 invoice, 1 customer, UK only, leakage £168.47K, rate 100%, net revenue £0.
Click United Kingdom	7,315 cancelled invoices, 2,307 customers, 2,685 products, gross £17,410.20K, leakage £1,266.21K, rate 7.27%.
Month = Dec, then United Kingdom	780 cancelled invoices, 552 customers, 901 products, leakage £297.98K, rate 14.29%. Without the £168.47K reversal, the rate is about 6.8%.